

Operating Systems – 3 easy pieces

Chap. 37: RAID

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**Dear Lord, please grant me
the ability to punch people
over the Internet.**



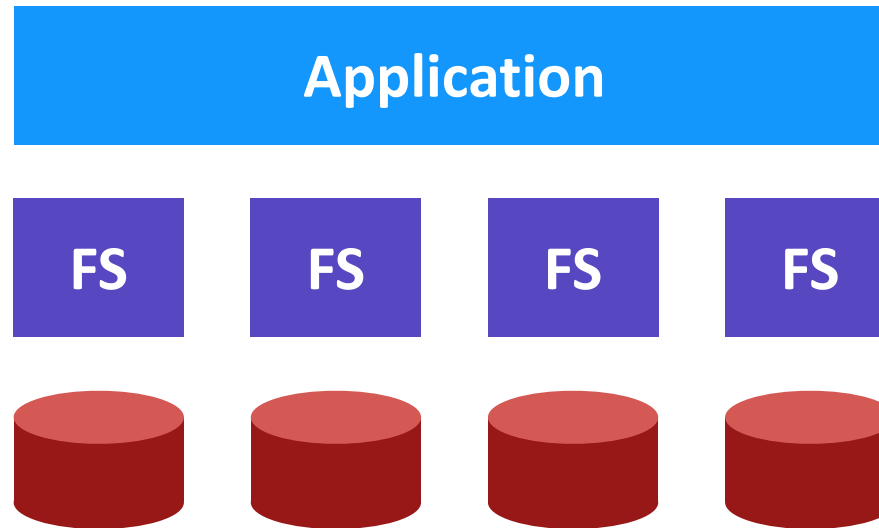
Only One Disk?

Sometimes we want many disks — why?

- Capacity
- Reliability
- Performance

Challenge: most file systems work on only one disk

Sol. 1: JBOD



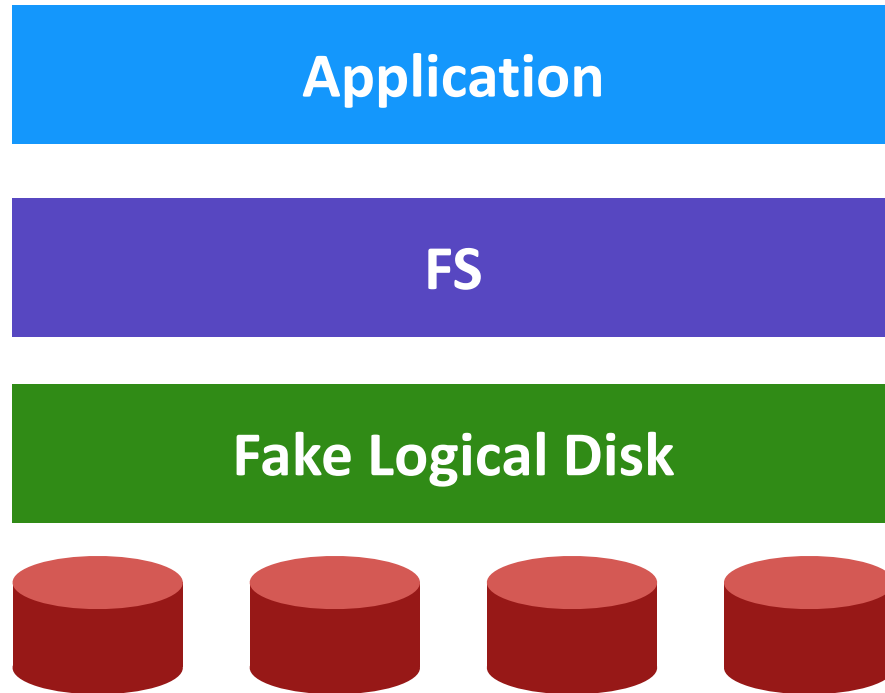
Application is smart, stores different files on different file systems.

JBOD: Just a Bunch Of Disks

Sol. 2: RAID

RAID is:

- Transparent
- Deployable

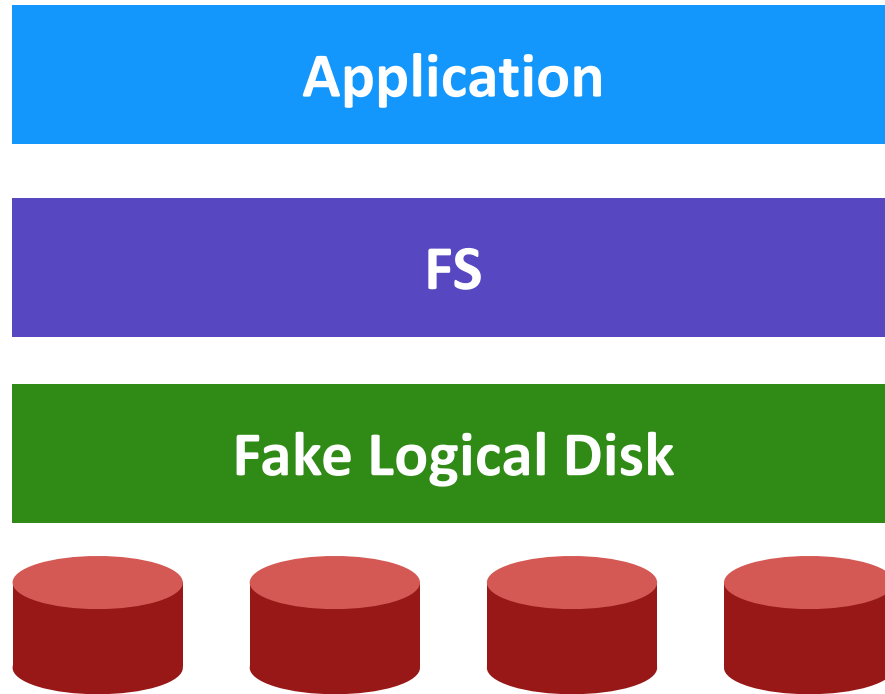


Logical disk gives

- Capacity
- Reliability
- Performance

Build logical disk from many physical disks.
RAID: **R**edundant **A**rray of **I**nexpensive **D**isks

Sol. 2: RAID



RAIDs can tolerate the loss of a disk and keep operating as if nothing were wrong.

RAID (Redundant Array of Inexpensive Disks)

- ❑ **Use multiple disks** in concert to build a **faster, bigger**, and more **reliable** disk system.
 - ◆ RAID just looks like a big disk to the host system.
- ❑ Advantage
 - ◆ **Performance & Capacity**: Using multiple disks in parallel
 - ◆ **Reliability**: RAID can tolerate the loss of a disk.

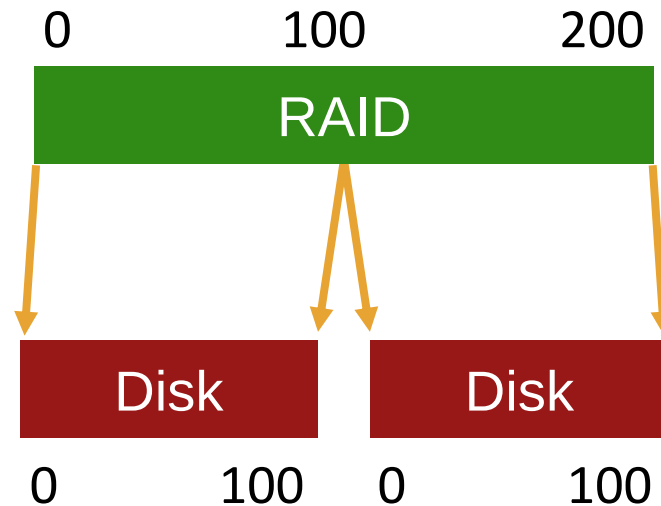
RAIDs provide these advantages **transparently** to systems that use them.

Note

RAID is a perfect example, and certainly its transparency contributed to its success; administrators could install a SCSI-based RAID storage array instead of a SCSI disk, and the rest of the system (host computer, OS, etc.) did not have to change one bit to start using it. **By solving this problem of deployment**, RAID was made more successful from day one.

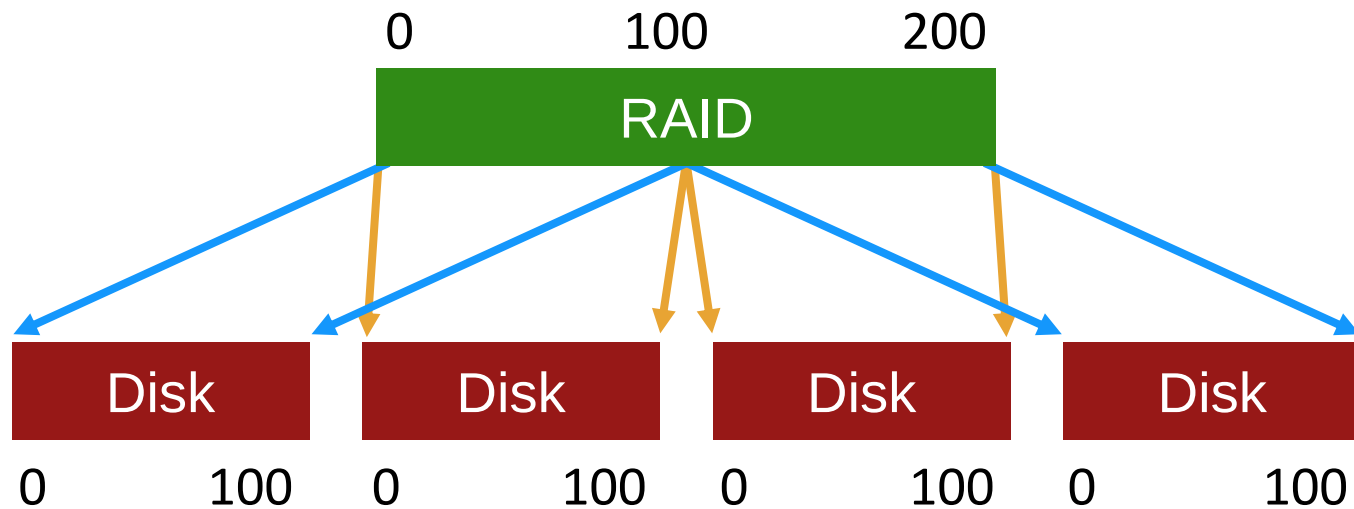
General Strategy: MAPPING

Build fast, large disk from smaller ones.



General Strategy: REDUNDANCY

Add even more disks for reliability.



- ▣ When a RAID receives I/O request,
 1. The RAID **calculates** which disk to access.
 2. The RAID **issue** one or more **physical I/Os** to do so.

- ▣ RAID example: A mirrored RAID system
 - ◆ Keep two copies of each block (each one on a separate disk)
 - ◆ Perform two physical I/Os for every one logical I/O it is issued.

RAID Internals

- ❑ A microcontroller:
 - ◆ Run firmware to direct the operation of the RAID
- ❑ Volatile memory (such as DRAM)
 - ◆ Buffer data blocks
- ❑ Non-volatile memory
 - ◆ Buffer writes safely
- ❑ Specialized logic to perform parity calculation

- ▣ RAIDs are designed to **detect** and **recover** from certain kinds of disk faults.
- ▣ **Fail-stop** fault model
 - ◆ A disk can be in one of two states: **Working** or **Failed**.
 - Working: all blocks can be read or written.
 - Failed: the disk is permanently lost.
 - ◆ RAID controller can immediately observe when a disk has failed.

How to evaluate a RAID

▣ Capacity

- ◆ How much useful capacity is available to systems?

▣ Reliability

- ◆ How many disk faults can the given design tolerate?

▣ Performance

N := number of disks

C := capacity of 1 disk

S := sequential throughput of 1 disk

R := random throughput of 1 disk

D := latency of one small I/O operation

▣ Capacity

- ◆ Given a set of N disks each with B blocks, how much useful capacity is available to clients of the RAID?
 - Without redundancy, the answer is $(N \times B)$;
 - With redundancy, if a system keeps two copies of each block (called **mirroring**), we obtain a useful capacity of $\frac{(N \times B)}{2}$.

RAID Levels

	RAID-0	RAID-1		RAID-4	RAID-5
Capacity	N	N/1		N-1	N-1
Reliability	0	1 (for sure) $\frac{N}{2}$ (if lucky)		1	1
Throughput					
Sequential Read	$N \cdot S$	$(N/2) \cdot S$	RAID Level 2	$(N-1) \cdot S$	$(N-1) \cdot S$
Sequential Write	$N \cdot S$	$(N/2) \cdot S$	RAID Level 3	$(N-1) \cdot S$	$(N-1) \cdot S$
Random Read	$N \cdot R$	$N \cdot R$		$(N-1) \cdot R$	$N \cdot R$
Random Write	$N \cdot R$	$(N/2) \cdot R$		$\frac{1}{2} R$	$\frac{N}{4} R$
Latency					
Read	D	D		D	D
Write	D	D		2D	2D
					RAID Level 6

N : the number of disks

D : the time that a request to a single disk take

RAID is also built as a software layer: **software RAID**

RAID Level 0: Striping

- ▣ RAID Level 0 is the simplest form as **striping** blocks.
 - ◆ **Spread the blocks** across the disks in a round-robin fashion.
 - ◆ No redundancy
 - ◆ Excellent performance and capacity

Disk 0	Disk 1	Disk 2	Disk 3
0	1	2	3
4	5	6	7
8	9	10	11
12	13	14	15

-----> Stripe
(The blocks in the same row)

RAID-0: Simple Striping
(Assume here a 4-disk array)

RAID Level 0 (Cont.)

- Example) RAID-0 with a bigger chunk size
 - ◆ Chunk size : 2 blocks (8 KB)
 - ◆ A Stripe: 4 chunks (32 KB)

Disk 0	Disk 1	Disk 2	Disk 3	
0	2	4	6	chunk size: 2blocks
1	3	5	7	
8	10	12	14	
9	11	13	15	

Striping with a Bigger Chunk Size

Chunk Sizes

- ▣ Chunk size mostly affects performance of the array
 - ◆ **Small chunk size**
 - Increasing the parallelism
 - Increasing positioning time to access blocks
 - ◆ **Big chunk size**
 - Reducing intra-file parallelism
 - Reducing positioning time

**Determining the “best” chunk size is hard to do.
Most arrays use larger chunk sizes (e.g., 64 KB)**

Mapping Problem

- Given a logical block address A , the RAID can easily compute the desired disk and offset with two simple equations:
 - $\text{Disk} = A \% \text{number of disks}$
 - $\text{Offset} = \frac{A}{\text{number of disks}}$
- Note: These are int operations, for example $\frac{4}{3} = 1$ not 1.33333..
- Example: Request arrives for block 14.
 - Given that there are 4 disks, this would mean that the disk we are interested in is ($14 \% 4 = 2$): disk 2.
 - The exact block is calculated as ($\frac{14}{4} = 3$): block 3.
 - Thus, block 14 should be found on the fourth block (block 3, starting at 0) of the third disk (disk 2, starting at 0), which is exactly where it is.

Mapping Problem

- Example: Request arrives for block 14.
 - Given that there are 4 disks, this would mean that the disk we are interested in is $(14 \% 4 = 2)$: disk 2.
 - The exact block is calculated as $\left(\frac{14}{4} = 3\right)$: block 3.
 - Thus, block 14 should be found on the fourth block (block 3, starting at 0) of the third disk (disk 2, starting at 0), which is exactly where it is.

Disk 0	Disk 1	Disk 2	Disk 3
0	1	2	3
4	5	6	7
8	9	10	11
12	13	14	15

Figure 38.1: **RAID-0: Simple Striping**

N : the number of disks

- ▣ **Capacity** → RAID-0 is perfect.
 - ◆ Striping delivers N disks worth of useful capacity.

- ▣ **Performance** of striping → RAID-0 is excellent.
 - ◆ All disks are utilized often in parallel.

- ▣ **Reliability** → RAID-0 is bad.
 - ◆ Any disk failure will lead to data loss.

Evaluating RAID Performance

- Consider two performance metrics
 - ◆ Single request latency
 - ◆ Steady-state throughput
- Workload
 - ◆ **Sequential**: access 1MB of data (block (B) ~ block (B + 1MB))
 - ◆ **Random**: access 4KB at random logical address
- A disk can transfer data at
 - ◆ S MB/s under a sequential workload
 - ◆ R MB/s under a random workload

Evaluating RAID Performance Example

▣ sequential (S) vs random (R)

- ◆ **Sequential** : transfer 10 MB on average as continuous data.
- ◆ **Random** : transfer 10 KB on average.
- ◆ Average seek time: 7 ms
- ◆ Average rotational delay: 3 ms
- ◆ Transfer rate of disk: 50 MB/s

▣ Results:

$$◆ S = \frac{\text{Amount of Data}}{\text{Time to access}} = \frac{10 \text{ MB}}{210 \text{ ms}} = 47.62 \text{ MB /s}$$

$$◆ R = \frac{\text{Amount of Data}}{\text{Time to access}} = \frac{10 \text{ KB}}{10.195 \text{ ms}} = 0.981 \text{ MB /s}$$

Evaluating RAID-0 Performance

N : the number of disks

- ▣ Single request latency
 - ◆ Identical to that of a single disk.

- ▣ Steady-state throughput
 - ◆ **Sequential** workload : $N \cdot S$ MB/s
 - ◆ **Random** workload : $N \cdot S$ MB /s



**“Dear Andy: How have you been?
Your mother and I are fine. We miss you.
Please sign off your computer and come
downstairs for something to eat. Love, Dad.”**

RAID Level 1 : Mirroring

- ▣ RAID Level 1 tolerates **disk failures**.
 - ◆ **Copy** more than one of **each block** in the system.
 - ◆ Copy block places on a separate disk.

Disk 0	Disk 1	Disk 2	Disk 3
0	0	1	1
2	2	3	3
4	4	5	5
6	6	7	7

Simple RAID-1: Mirroring (Keep two physical copies)

N : the number of disks

- **Capacity:** RAID-1 is Expensive

- ◆ The useful capacity of RAID-1 is $N/2$.

- **Reliability:** RAID-1 does well.

- ◆ It can tolerate the failure of any one disk (up to $N/2$ failures depending on which disk fail).

Performance of RAID-1

□ Two physical writes to complete

- ◆ It suffers the worst-case seek and rotational delay of the two request.
- ◆ Steady-state throughput

- **Sequential Write** : $\frac{N}{2} \cdot S$ MB/s

- Each logical write must result in two physical writes.

- **Sequential Read** : $\frac{N}{2} \cdot S$ MB/s

- Each disk will only deliver half its peak bandwidth.

- **Random Write** : $\frac{N}{2} \cdot R$ MB/s

- Each logical write must turn into two physical writes.

- **Random Read** : $N \cdot R$ MB/s

- Distribute the reads across all the disks.

RAID Level 4 : Saving Space With Parity

- Add a single parity block

- ◆ A Parity block stores the *redundant information* for that stripe of blocks.

* P: Parity

Disk 0	Disk 1	Disk 2	Disk 3	Disk 4
0	0	1	1	P0
2	2	3	3	P1
4	4	5	5	P2
6	6	7	7	P3

Five-disk RAID-4 system layout

RAID Level 4 (Cont.)

- **Compute parity** : the XOR of all of bits

C0	C1	C2	C3	P
0	0	1	1	$\text{XOR}(0,0,1,1)=0$
0	1	0	0	$\text{XOR}(0,1,0,0)=1$

- **Recover from parity**

- ◆ Imagine the bit of the C2 in the first row is lost.
 1. Reading the other values in that row : 0, 0, 1
 2. The parity bit is 0 → even number of 1's in the row
 3. What the missing data must be: a 1.

N : the number of disks

□ Capacity

- ◆ The useful capacity is $(N - 1)$.

□ Reliability

- ◆ RAID-4 tolerates 1 disk failure and no more.

▣ Performance

- ◆ Steady-state throughput
 - Sequential read: $(N - 1) \cdot S$ MB/s
 - Sequential write: $(N - 1) \cdot S$ MB/s

Disk 0	Disk 1	Disk 2	Disk 3	Disk 4
0	1	2	3	P0
4	5	6	7	P1
8	9	10	11	P2
12	13	14	15	P3

Full-stripe Writes In RAID-4

- Random read: $(N - 1) \cdot R$ MB/s

Random write performance for RAID-4

- Overwrite a block + update the parity
- **Method 1:** *additive parity*
 - ◆ Read in all of the other data blocks in the stripe
 - ◆ XOR those blocks with the new block (1)
 - ◆ **Problem:** the performance scales with the number of disks

Random write performance for RAID-4 (Cont.)

▣ Method 2: *subtractive parity*

C0	C1	C2	C3	P
0	0	1	1	$XOR(0,0,1,1)=0$

◆ Update C2(old) → C2(new)

1. Read in the old data at C2 (C2(old)=1) and the old parity (P(old)=0)
2. Calculate P(new): $P(new) = (C2(old) XOR C2(new)) XOR P(old)$
 - If C2(new)=C2(old) → P(new)=P(old)
 - If C2(new)≠C2(old) → Flip the old parity bit

Block0	Block1	Block2	Block3	Parity
00	10	11	10	11
10	01	00	01	10

Small-write problem

- The parity disk can be a **bottleneck**.
 - ◆ Example: update blocks 4 and 13 (marked with *)

Disk 0	Disk 1	Disk 2	Disk 3	Disk 4
0	1	2	3	P0
*4	5	6	7	+P1
8	9	10	11	P2
12	*13	14	15	+P3

Writes To 4, 13 And Respective Parity Blocks.

- Disk 0 and Disk 1 can be accessed in parallel.
- Disk 4 prevents any parallelism.

RAID-4 throughput under random small writes is $\left(\frac{R}{2}\right)$ MB/s (*terrible*).

A I/O latency in RAID-4

▣ A single read

- ◆ Equivalent to the latency of a single disk request.

▣ A single write

- ◆ Two reads and then two writes
 - Data block + Parity block
 - The reads and writes can happen in parallel.
- ◆ Total latency *is about twice* that of a single disk.



"IF AT FIRST YOU DON'T SUCCEED, BLAME YOUR COMPUTER."

RAID Level 5: Rotating Parity

- ▣ RAID-5 is **solution of** small write problem.
 - ◆ Rotate the parity blocks across drives.
 - ◆ Remove the parity-disk bottleneck for RAID-4

Disk 0	Disk 1	Disk 2	Disk 3	Disk 4
0	1	2	3	P0
5	6	7	P1	4
10	11	P2	8	9
15	P3	12	13	14
P4	16	17	18	19

RAID-5 With Rotated Parity

N : the number of disks

□ Capacity

- ◆ The useful capacity for a RAID group is $(N - 1)$.

□ Reliability

- ◆ RAID-5 tolerates 1 disk failure and no more.

N : the number of disks

▣ Performance

- ◆ Sequential read and write
 - ◆ A single read and write request
- } Same as RAID-4
-
- ◆ Random read : a little better than RAID-4
 - RAID-5 can utilize all of the disks.
 - ◆ Random write : $\frac{N}{4} \cdot R$ MB/s
 - The factor of four loss is cost of using parity-based RAID.

RAID Comparison: A Summary

N : the number of disks

D : the time that a request to a single disk take

	RAID-0	RAID-1	RAID-4	RAID-5
Capacity	N	$N/1$	$N-1$	$N-1$
Reliability	0	1 (for sure) $\frac{N}{2}$ (if lucky)	1	1
Throughput				
Sequential Read	$N \cdot S$	$(N/2) \cdot S$	$(N-1) \cdot S$	$(N-1) \cdot S$
Sequential Write	$N \cdot S$	$(N/2) \cdot S$	$(N-1) \cdot S$	$(N-1) \cdot S$
Random Read	$N \cdot R$	$N \cdot R$	$(N-1) \cdot R$	$N \cdot R$
Random Write	$N \cdot R$	$(N/2) \cdot R$	$\frac{1}{2} R$	$\frac{N}{4} R$
Latency				
Read	D	D	D	D
Write	D	D	$2D$	$2D$

RAID Capacity, Reliability, and Performance

RAID Comparison: A Summary

- ▣ **Performance** and do not care about reliability → RAID-0 (Striping)
- ▣ **Random I/O** performance and **Reliability** → RAID-1 (Mirroring)
- ▣ **Capacity** and **Reliability** → RAID-5
- ▣ **Sequential I/O** and Maximize **Capacity** → RAID-5

Conclusion



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